problems/MHD/kelvinHelmholtzChan2022

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Summary. This case uses **neither DynSGS nor any positivity mechanism**. Its deck selects `:visc_model => SMAG()` (Smagorinsky); `:lpositivity` is left at `false`, the solver carries no stage limiter and the flux has no floor. It is included in this set because it lives in `problems/MHD`; the stabilization it does use is described below so that the comparison with the eight `DSGS_MHD` cases is complete.

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	Chan et al. (2022) §3.2.1
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 5/3$	`user_flux.jl`
Domain, mesh	doubly periodic 32×32 quads, `./meshes/gmsh_grids/hexa_TFI_32x32_unitsquare.msh`	the file is not in this repository (JexpressoMeshes)
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 7$	
Time integration	`CarpenterKennedy2N54`, fixed Δt, $\Delta t = 2 \times 10^{-3}$, $t_{end} = 15$	
Variables	`TOTAL()`, `"energy"`	
Initial data	$\tilde{B} = \tanh(15y + 7.5) - \tanh(15y - 7.5)$, $\rho = \frac{1}{2} + \frac{3}{4}\tilde{B}$, $p = 1$, $u = \frac{1}{2}(\tilde{B} - 1)$, $v = \frac{1}{10}\sin 2\pi x$, $\mathbf{B} = (0, 0.125, 0)$	`initialize.jl`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; $S_\psi = -(c_h/0.18)\,\psi$	
Sources	`:lsource => true` (ψ damping only)	
Boundary conditions	periodic; free-slip fallback only for non-periodic tags	`user_bc.jl`
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 Stabilization actually used: Smagorinsky (not DynSGS)

Deck key / item	Value as run	Source / remark
`:visc_model`	`SMAG()`	**not** `DSGS_MHD()`; no `:dsgs_*` key is read

Deck key / item	Value as run	Source / remark
`:lvisc`	true	
`:`μ	$(0, 1, 1, 1, 1, 1, 1, 1, 1)$	no mass diffusion (all `DSGS_MHD` cases have $\mu[1] = 1$)
Smagorinsky constants	$C_s = 0.21$, $\mathrm{Pr}_t = 0.7$, $\mathrm{Sc}_t = 0.7$, $\mu_{mol} = 1.8 \times 10^{-5}$, $\kappa_{mol} = 2.4 \times 10^{-5}$	`PhysicalConst` (SI air values, used as is in this nondimensional case)
Filter length	$\Delta = \Delta^{elem}_{\max}/N$ (`mesh.Δeffective_l`)	one value for the whole mesh

Per LGL node (`compute_sgs_cache!(::SGS_SMAG)`, `SGS.jl:3343`; reader `SGS.jl:3590`):

$$\mu_t = \rho\, C_s^2 \Delta^2 |S|, \qquad |S| = \sqrt{2 S_{ij} S_{ij}}, \quad S_{ij} = \tfrac{1}{2}(\partial_j u_i + \partial_i u_j), \; i, j \in \{x, y\},$$

no Richardson factor (`:lrichardson => false`). Slot coefficients, with the primitives $(\rho, u, v, T, w, \mathbf{B}, \psi)$ of `user_primitives.jl`:

Equation	Flux	Coefficient
ρ	—	$\mu[1] = 0$
$\rho u, \rho v$	deviatoric stress $\boldsymbol{\tau}$ of (u, v)	$\mu_{mol} + \mu_t$
E	$\kappa \nabla T + \boldsymbol{\tau} \cdot \mathbf{v}$	$\kappa = \kappa_{mol} + \mu_t/(\rho\, \mathrm{Pr}_t)$
ρw, B_k, ψ	$\kappa_s \nabla(w,\ B_k,\ \psi)$	$\kappa_s = \kappa_{mol} + \mu_t/(\rho\, \mathrm{Sc}_t)$

No resistive work is added to E on this path. The coefficient is blind to resolution: it is set by the strain rate, not by a residual.

3 Positivity

- `:lpositivity` false; no stage limiter; no state modification.
- No flux-only floor (`pressure_mhd` unguarded).
- No mass diffusion ($\mu[1] = 0$).

4 Observations

1. The mesh named in the deck is not shipped in `problems/MHD/kelvinHelmholtzChan2022` nor under `meshes/gmsh_grids/` in this checkout; only `KHI_32x32_periodic.geo` is.
2. μ_{mol} and κ_{mol} are `PhysicalConst`'s SI values for air; in this nondimensional setup they act as small constant background coefficients.

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	Where (file:line)
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report `:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:456-477` `src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`

Mechanism	**Where (file:line)**
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs}+2$ in 2D, $n_{eqs}+1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty $\Rightarrow$ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)